

In this course, I gained valuable experience in developing scalable, secure, and thoroughly tested software systems. I learned to integrate key backend technologies such as FastAPI, SQLAlchemy, and Pytest, using them to manage databases and build reliable testing suites. The course emphasized best practices in writing clean, modular, and maintainable code.

The User Management System project was especially impactful, helping me strengthen my skills in designing RESTful APIs, implementing user authentication and authorization with OAuth2, and validating data to ensure integrity. One of the highlights was developing advanced features like email notifications, role-based access control, and user status upgrades, all while maintaining secure and efficient database interactions.

Through writing over ten test cases, I honed my ability to debug and cover edge cases effectively. I also became more confident in resolving merge conflicts, addressing CI/CD pipeline issues, and deploying applications to DockerHub—deepening my understanding of the software development lifecycle and DevOps practices.

This project pushed me to think analytically, collaborate with others, and apply theoretical knowledge in a real-world setting. It gave me the tools to approach complex software challenges with professionalism and confidence.

Links:

[Issue 1](#)

[Issue 2](#)

[Issue 3](#)

[Issue 4](#)

[Issue 5](#)

[Tests](#) – Note: Not all tests are in this branch, I updated it according to each branch after a while since I forgot to switch in between branches.

[Feature](#)